

Valerio Paventi

XR & Real-Time 3D Engineer

Rome, Italy | valeriopaventi@gmail.com | [LinkedIn](#) | [Portfolio](#)

Available immediately | Open to remote roles across the EU

PROFILE

XR and real-time 3D engineer with 9 years of Unity experience across mobile AR and standalone VR. Shipped a commercial VR title on Meta Quest and PlayStation VR2, managing the PS5 port through Sony TRC certification, and delivered mobile AR products for education, retail and advertising as CTO for five years. Strong background in performance optimisation on constrained hardware, data-oriented design (ECS/DOTS) and technical leadership of small teams.

TECHNICAL SKILLS

XR & AR: ARKit, ARCore, Meta Quest, PlayStation VR2, standalone VR development, passthrough / mixed reality, plane and image tracking, VR interaction and comfort design

Real-Time 3D: Unity3D (URP, custom shaders, VFX, lightmapping, profiling), Unreal Engine, physics simulation, LOD and impostor systems

Performance: GPU and CPU profiling on mobile-class hardware, draw call and memory optimisation, data-oriented design (ECS/DOTS, Job System, Burst)

Platforms & Release: Meta Store, PlayStation Store (Sony TRC certification), iOS (Xcode), Android (Android Studio), WebGL

Languages: C#, C++, Python

Backend & Infra: .NET, REST APIs, MongoDB, AWS, Mirror Networking, Git

PROFESSIONAL EXPERIENCE

Taco Studios — XR & Gameplay Engineer

SEPTEMBER 2022 – PRESENT · ROME, ITALY

Mars Kitchen / VR Simulation · Unity · Meta Quest, PlayStation VR2 — Released

- Shipped a commercial VR title on Meta Quest and PlayStation VR2, working within the frame-rate and memory budgets of standalone headsets
- Architected a runtime lightmap swapping system enabling instant environment changes without scene reloads, preserving immersion under strict memory constraints
- Engineered an AI virtualisation system simulating off-screen NPC behaviour, maintaining complex simulation state at minimal CPU cost
- Built a custom impostor system for distant geometry, recovering significant frame budget on standalone hardware
- Tuned rigid-body physics for stable, responsive hand interaction — a core requirement for comfort in VR
- Authored custom shaders and particle systems optimised for mobile-class GPUs
- Led the PlayStation 5 / PSVR2 port: integrated hardware-specific features and PSN services, and passed Sony TRC certification
- Continuous profiling and debugging to hold target frame rates and physics stability across both platforms

Unreleased Title / Unity ECS/DOTS — Unreleased

- Architected data-oriented simulation systems using Unity ECS, the Job System and the Burst compiler
- Implemented deterministic procedural environment generation and streaming for large worlds

Polyverse & Acryptia / Unity · WebGL · real-time multiplayer

- Implemented systems inside a server-authoritative network architecture (Mirror), ensuring consistent state across clients
- Wrote a custom deterministic A* pathfinding algorithm and behaviour tree system from scratch to guarantee perfectly synchronised agent movement and eliminate desync
- Integrated modular inventory systems with blockchain smart contracts for verifiable asset ownership

Rainstones Srl — Chief Technology Officer — Mobile AR

JUNE 2019 – AUGUST 2022 · ROME, ITALY

- Led technical direction and a cross-functional team delivering mobile AR products for the education, retail and advertising sectors
- Architected scalable Unity AR clients on ARKit and ARCore, optimised to run across a broad range of mid-range mobile devices
- ARgo — lead developer: AR tracking and placement, UI/UX, content pipeline and release management
- Inquadra — iOS AR development with ARKit, including native integration through Xcode
- Lisciani — cross-platform AR experiences for educational products (ARKit / ARCore)
- Owned architecture decisions, technical leadership, testing and release processes
- Client work delivered under confidentiality; visual assets not shown

JLab Srls — Founder & Chief Technology Officer — AR

MAY 2017 – MAY 2019 · ROME, ITALY

- Co-founded the company and led its technical side, building and managing a small engineering team
- ExpertAR — lead developer across the full stack: ARKit / ARCore client, native Android development and backend services
- Owned the complete workflow, from ideation and design through frontend and backend development to release

EDUCATION & TRAINING

Liceo Scientifico Keplero — Scientific High School Diploma

Rome, 2009 – 2013

CEFI — Java Development course, 2015

Epic Games — Unreal Engine Fundamentals, 2025

Epic Games — Introduction to Game Design, 2025

LANGUAGES

Italian — native

English — B2, professional working proficiency

Spanish — A2, elementary